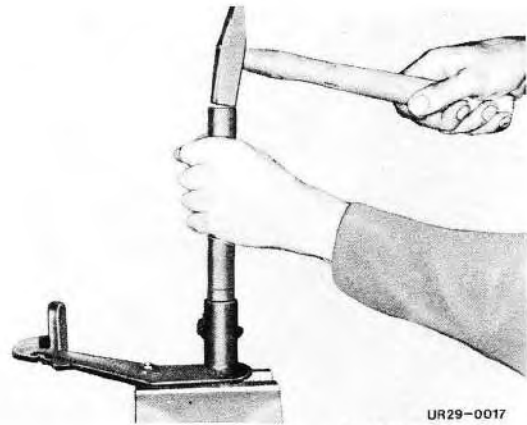


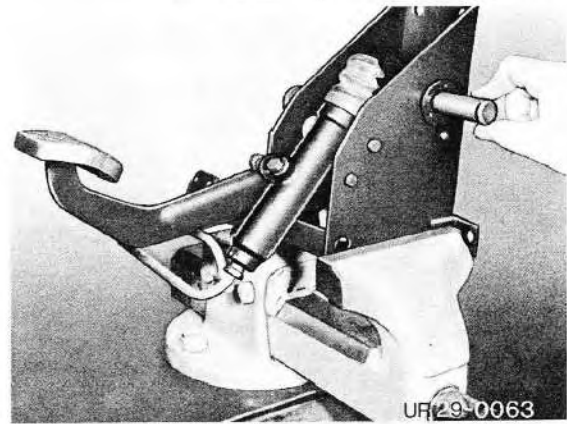
Installation

1 Force both needle bearings and spacing tube into clutch pedal.



UR29-0017

2 Fit bearing pin.

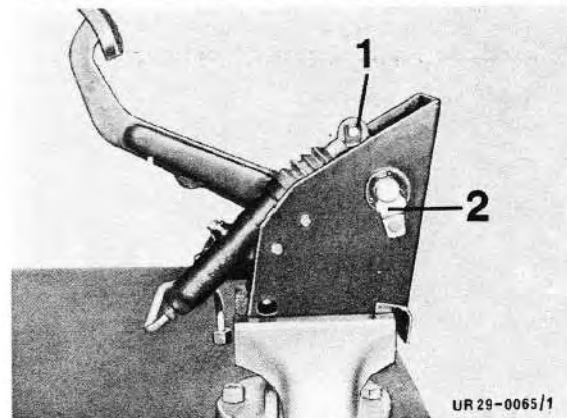


UR29-0063

3 Secure bearing pin (2) and push rod (1) of clutch master cylinder.

4 Force clutch pedal upwards.

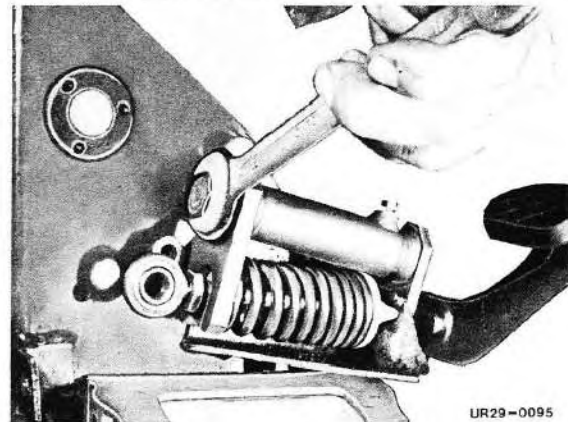
5 Insert needle bearing into push rod, using multipurpose grease.



UR 29-0065/1

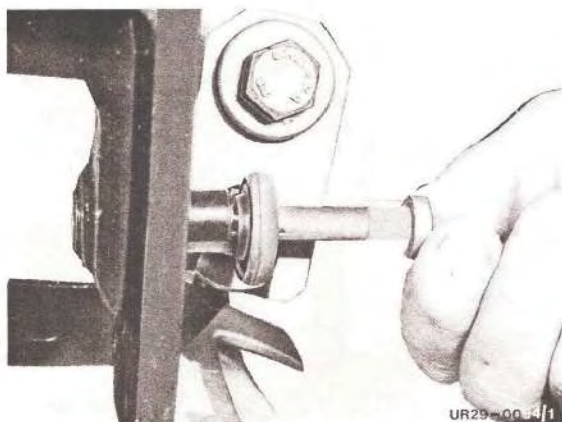
6 Slip upper spring seat, spring and lower spring seat onto push rod, installing with special tool No. 2.

Note: Spring must be fitted in such a way that end with closer turns is pointing toward spring mounting.

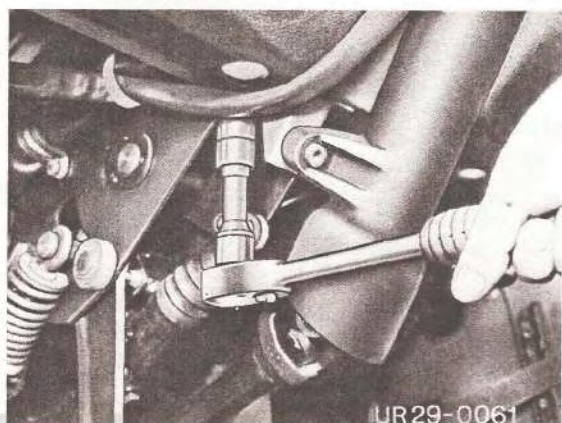


UR29-0095

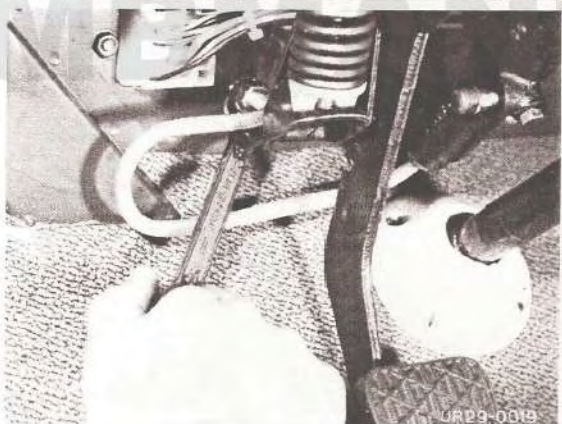
SA 35 737



- 7 Coat retaining pin with multipurpose grease, install and secure.



- 8 Bolt pedal bearing bracket on.
9 Attach wiring harness to pedal bearing bracket.



- 10 Attach pipe to connector.
11 Attach line to clutch master cylinder.
12 Adjust free travel, see 3.2/1.
Bleed hydraulic clutch operation, see 4.1/1.

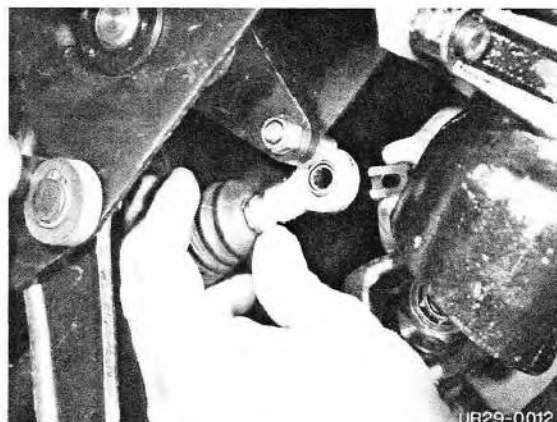
Note: Upon completion of installation, adjust spring at adjusting nut on push rod so that clutch pedal returns automatically to stop.

Removal and installation

CAUTION: DANGER

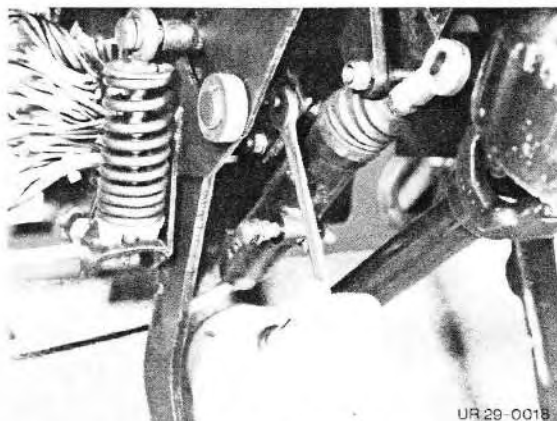
If mounted top dead center spring, watch for sudden dropping of clutch pedal.

1 Detach retainer holding clutch pedal, remove washer, release and unhook push rod of clutch master cylinder.



2 Unscrew clutch master cylinder at bracket.

3 Detach line at clutch master cylinder.



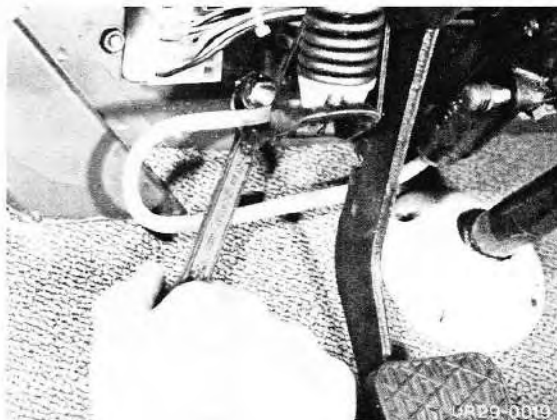
4 Unscrew pipe from connector.

5 Withdraw clutch master cylinder and pipe, unscrewing pipeline.

6 Clean and check all parts, exchanging if necessary.

7 Installation in opposite sequence to removal.

8 Adjust clutch master cylinder.
See 3.3/1.



Adjustment

1 Remove dust cap and release piston rod lock-nut.

2 Unscrew piston rod until it touches piston without pre-loading.

Note: Make sure that master cylinder piston is at neutral (stop on snap ring).

3 Now screw piston rod back into clevis or joint end by max. 1/4 turn.

4 Hold and secure piston rod, remembering to fit dust cap.

Note: Play of 0 = free play cannot be measured and has to be "felt" at clutch pedal. Complaints about delayed separation and poor clutch control are to be rectified by making the adjustment according to 3.3/1.



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Basic adjustment

1 Unscrew at union the hydraulic line to slave cylinder and collect brake fluid.

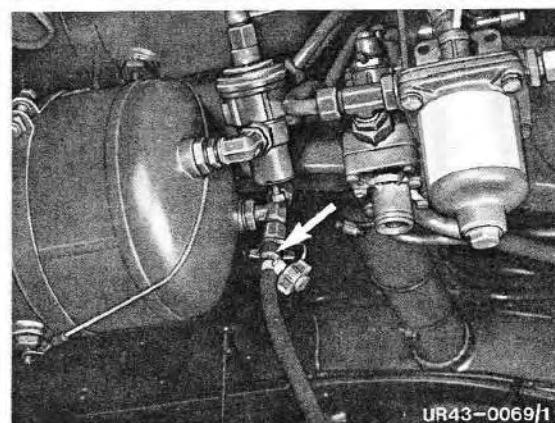
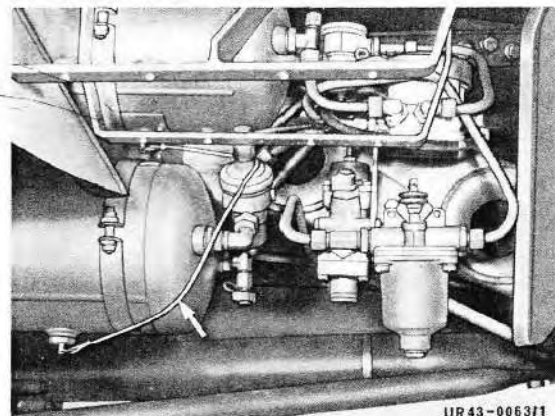
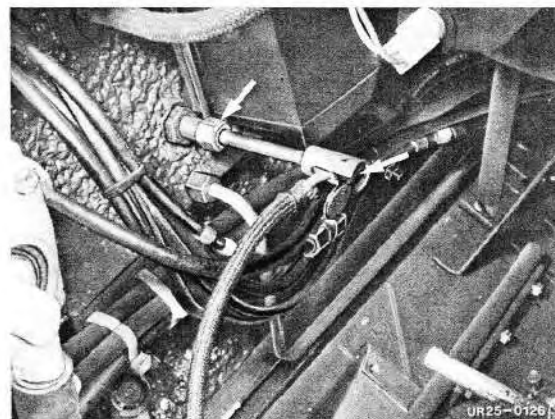
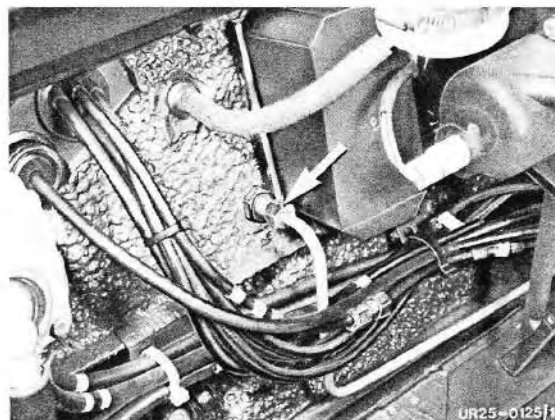
2 Empty brake fluid reservoir and connecting hose by pumping clutch pedal several times.

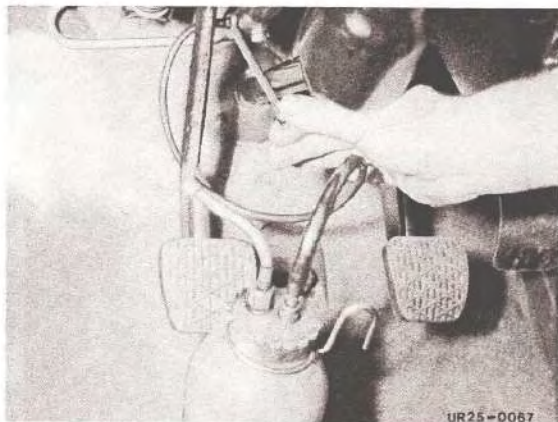
3 Attach special tool No. 1 to union, fitting tire inflating hose.

4 Lower air tank pressure to about 2 bar.

Note: Note pressure reading on cluster instrument in cab.

5 Screw tire inflating hose onto tire inflating or test connection.



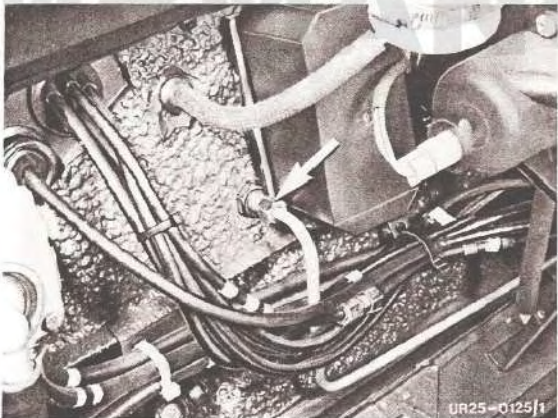


6 Open bleeder plug at master cylinder. Fit bleeder hose and immerse other end in vessel of brake fluid.



7 Release lock-nut on thrust rod and screw thrust rod out of clevis until no more air emerges from bleeder hose.

8 Screw thrust rod back into clevis until air begins to flow out of bleeder hose again. Now screw thrust rod into clevis by **1 more turn** and tighten lock-nut.



9 Detach tire inflating hose and special tool No.1. Attach hydraulic line to union.

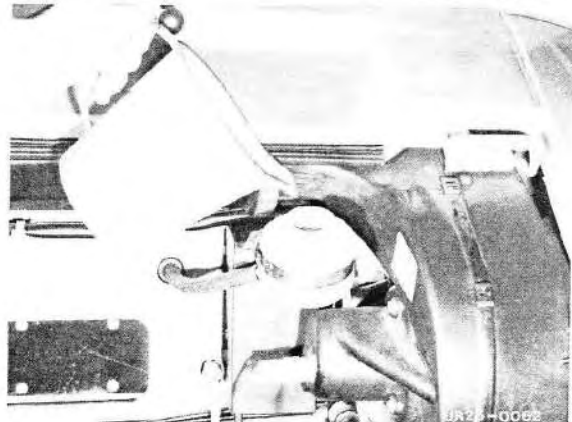
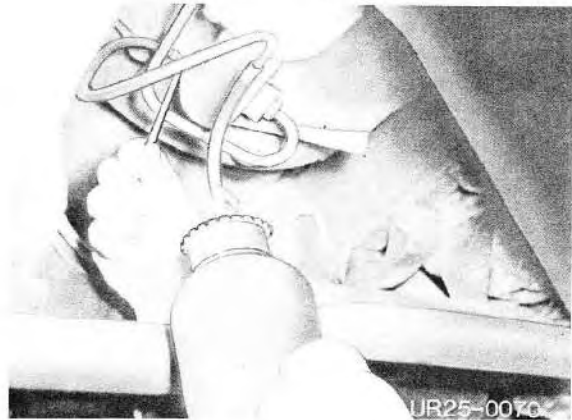
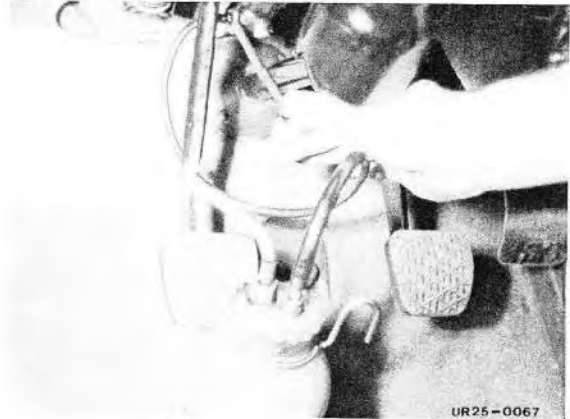
10 Bleed clutch actuating, referring to 4.1/1.

Bleeding

CAUTION: DANGER

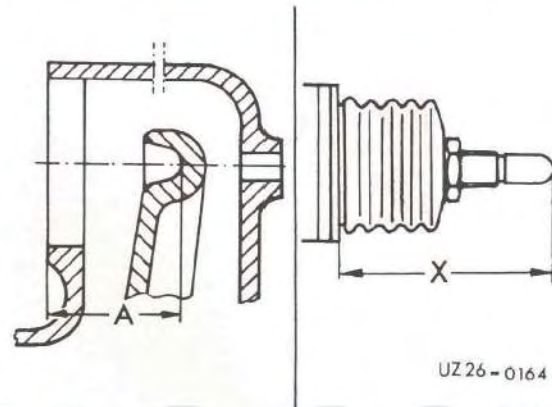
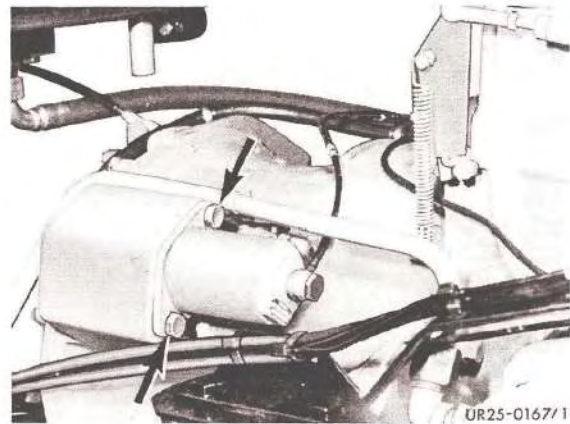
If mounted top dead center spring, watch for sudden dropping of clutch pedal.

- 1 Remove engine hood outside.
- 2 Prepare bleeder for use, referring to maker's instructions.
- 3 Unscrew reservoir cap and connect bleeder.
- 4 Detach dust cap from bleeder screw of master cylinder and connect bleeder bottle.
- 5 Open bleeder screw and allow brake fluid to run out until all air bubbles have disappeared.
- 6 Close bleeder screw. Disconnect bleeder bottle and fit dust cap.
- 7 Slave cylinder is to be bled in same sequence as described in operations 4 to 6.
- 8 Detach bleeder and correct level of brake fluid.
- 9 Screw cap onto reservoir.



Adjusting

- 1 Tilt cab, see 60.5-2.1/1 or 60.5-2.2/1.
- 2 Unbolt pto slave cylinder and remove.
- 3 Attach release lever to clutch and measure distance **A** from cylinder support to locating hollow.
- 4 Add the constant **16** to value **A** and adjust the piston travel to $X = A + 16$.
- 5 Lower cab, see 60.5-2.1/1 or 60.5-2.2/1.



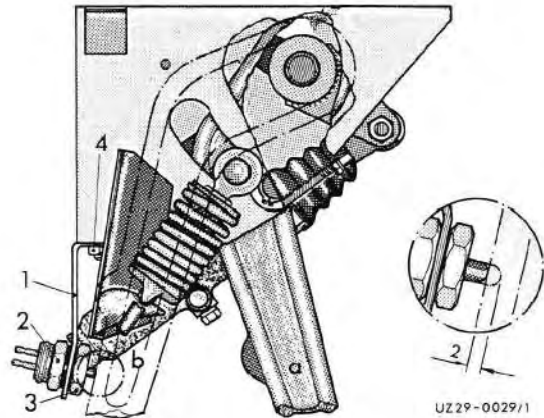
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Adjusting

- 1 Depress clutch pedal as far as it will go.
- 2 In this position **b** the plunger of the pressure switch must be pressed **2 mm**.

Note: Adjustment can be made by slackening the adjusting bolt on the bracket.

- a** Clutch pedal not operated
 - b** Clutch pedal at stop
 - 1** Bracket for pressure switch
 - 2** Pressure switch for starter interlock
 - 3** Mount for pressure switch
 - 4** Adjusting bolt
- 3 Check function of starter interlock.



Contents

Chapter	Page
1 General	
Installation survey	1.1/1
General view	1.1/1
Exploded view	1.2/1
2 Brake pedal for service brakes	
Removal and installation, with control cable	2.1/1
Removal and installation, with linkage	2.2/1

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Installation Survey

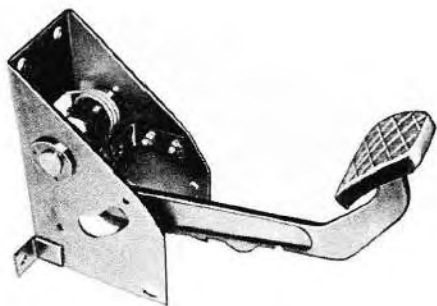
Chassis			Cab	Brake pedal Installation as of chassis end No.	
Model	Sales designation	Series Chassis end No.	Model	With control cable	With linkage
435.110	U 1700	from 002 230	425.800 425.820 425.821 425.825	to 002 099	from 002 100 ²⁾
.111	U 1700 L ¹⁾	from 000 008			
.113	U 1700 L/38	from 084 642			
.115	U 1300 L ¹⁾	from 000 019			
.117	U 1300 L/37	from 071 796			
.170	Attachment Condor	from PB ³⁾	—		

1) L = Truck

2) Chassis Model 435.115, special vehicle BW from chassis end No. 000 061

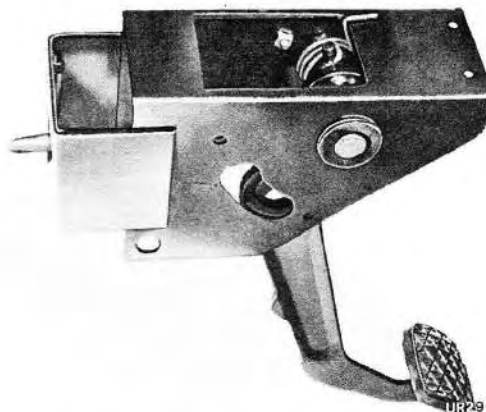
3) PB = Start of production

General view



UR29-0069

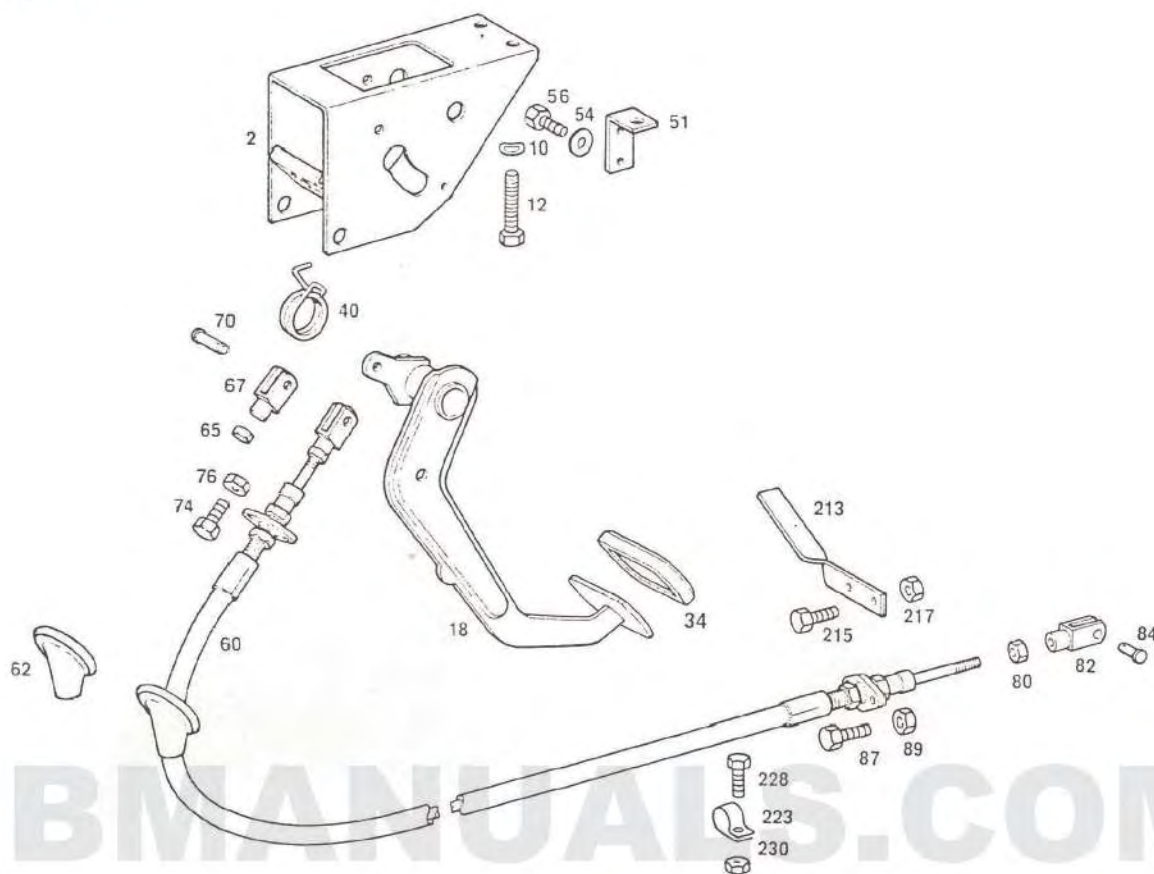
Brake pedal with cable



UR29-0096

Brake pedal with linkage

Exploded view



UZ29 - 0022

Brake pedal gear

- 2 Bearing bracket for brake pedal
- 10 Spring washer
- 12 Bolt
- 18 Pedal
- 34 Cap
- 40 Spring
- 51 Stop
- 54 Washer
- 56 Bolt
- 60 Control cable
- 62 Grommet
- 65 Nut
- 67 Clevis
- 70 Pin

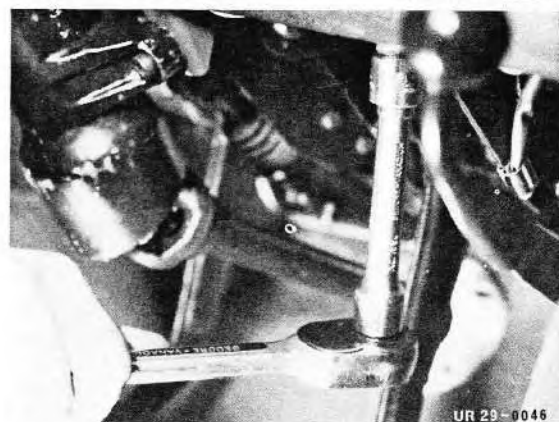
- 74 Bolt
- 76 Nut
- 80 Nut
- 82 Clevis
- 84 Pin
- 87 Bolt
- 89 Nut
- 213 Bracket
- 215 Bolt
- 217 Nut
- 223 Clamp
- 228 Bolt
- 230 Nut

Removal, with control cable

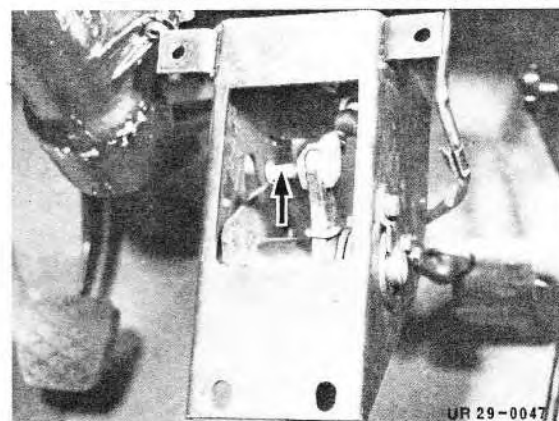
- 1 Unscrew cable from stoplight switch.



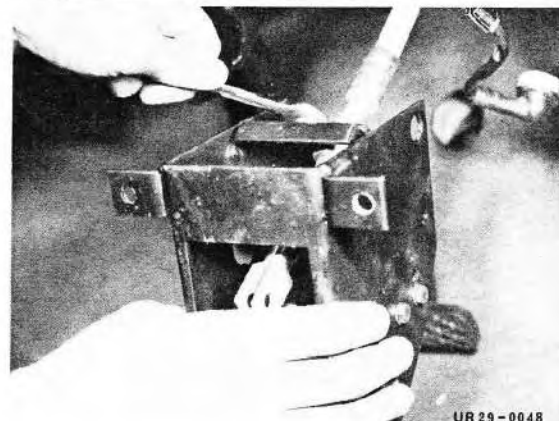
- 2 Unscrew complete pedal bearing bracket.

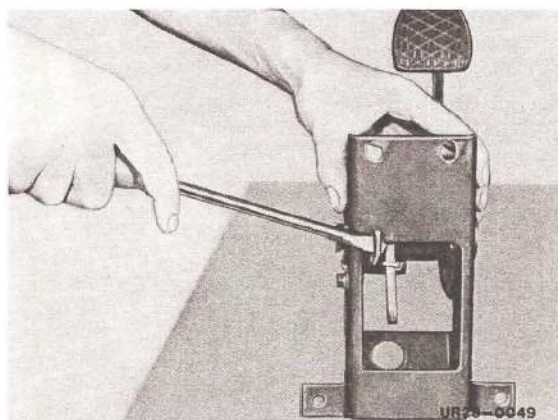


- 3 Release and remove pin from brake cable.

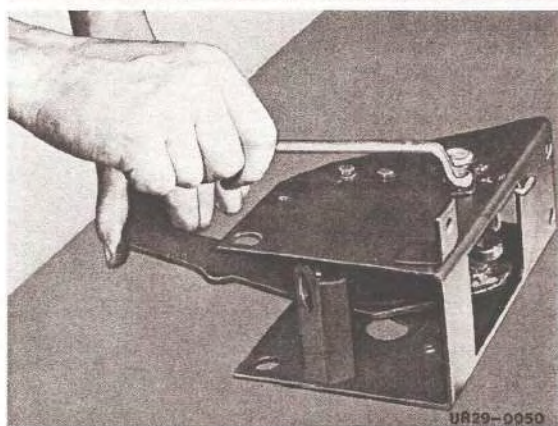


- 4 Remove brake cable with bearing ring and remove pedal bearing bracket.





5 Detach return spring.

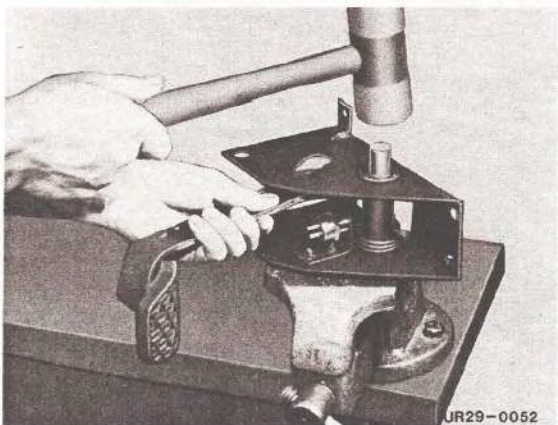


6 Unscrew retainer holding brake pedal bearing pin.



7 Force bearing pin out and detach brake pedal.

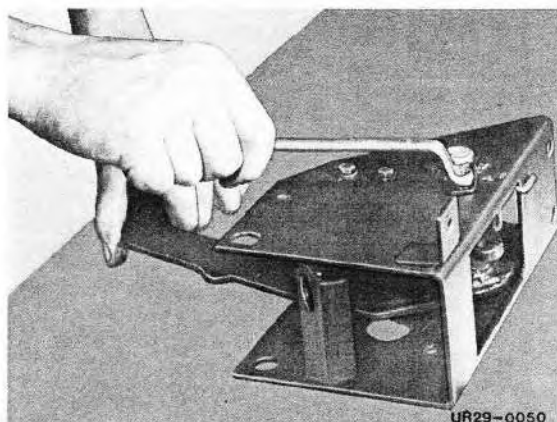
8 Clean and check all parts, exchanging if necessary.



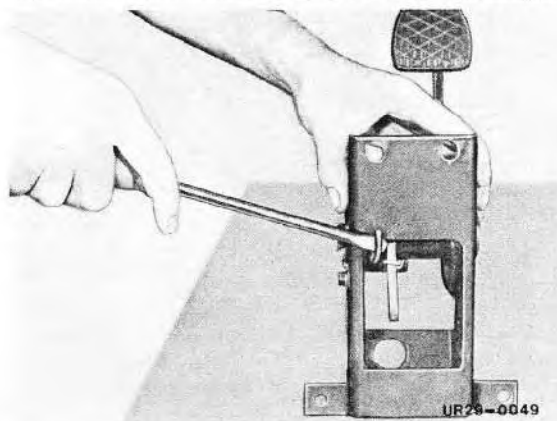
Installation, with control cable

1 Introduce new brake pedal into bearing bracket. Grease and insert bearing pin.

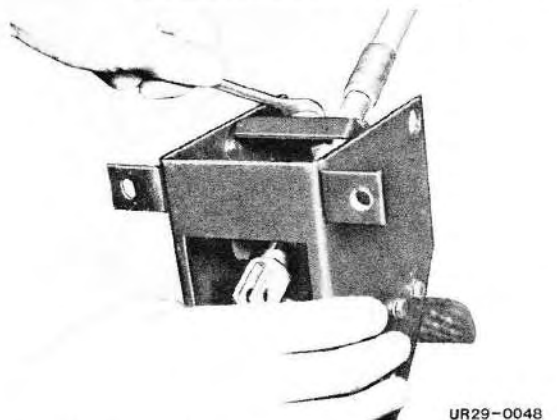
- 2 Screw bearing pin retainer on.



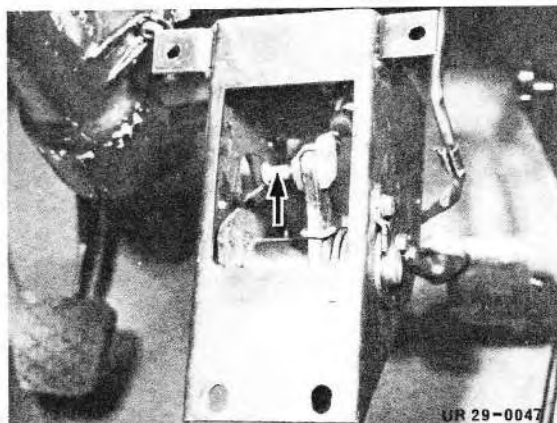
- 3 Attach return spring to brake pedal and pedal bearing bracket.

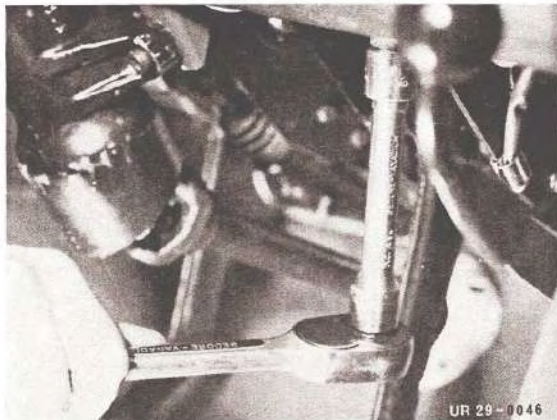


- 4 Attach brake rope and bearing ring to pedal bearing bracket.



- 5 Hook brake rope and pin onto brake pedal and insert cotter.





- 6 Bolt pedal bearing bracket on.

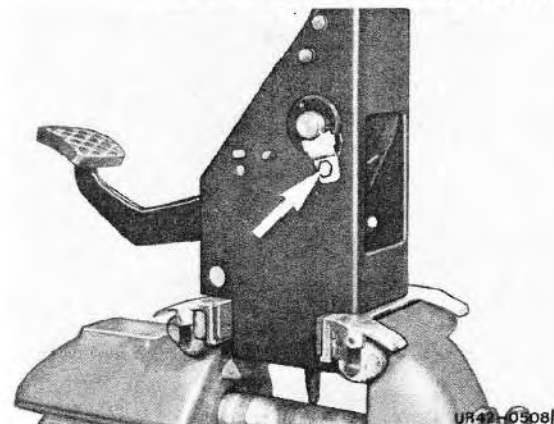
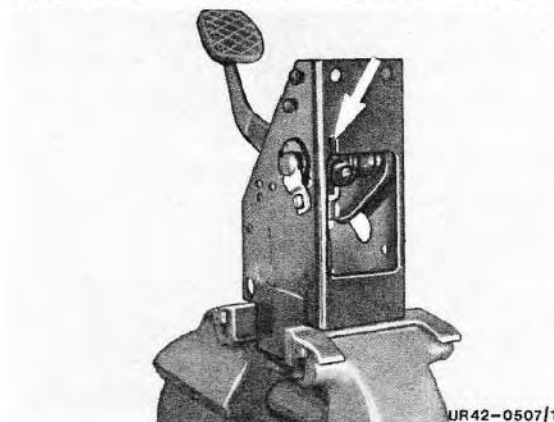
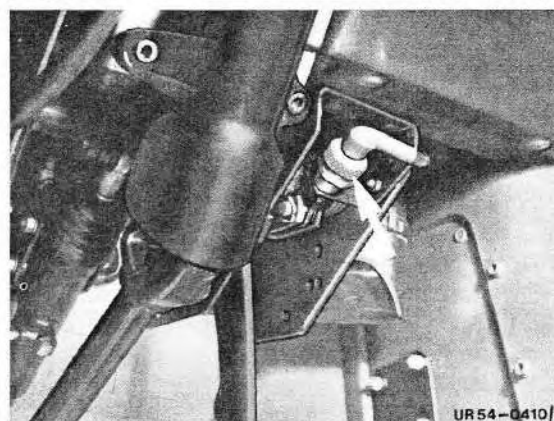
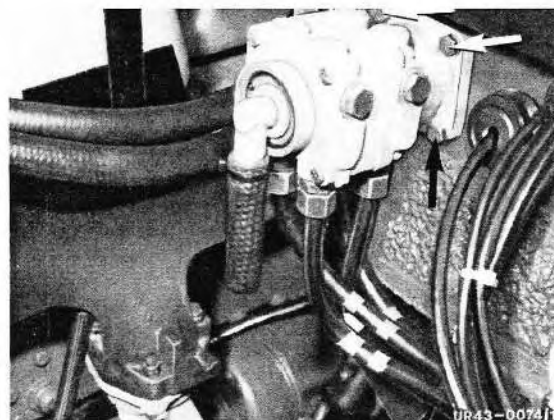


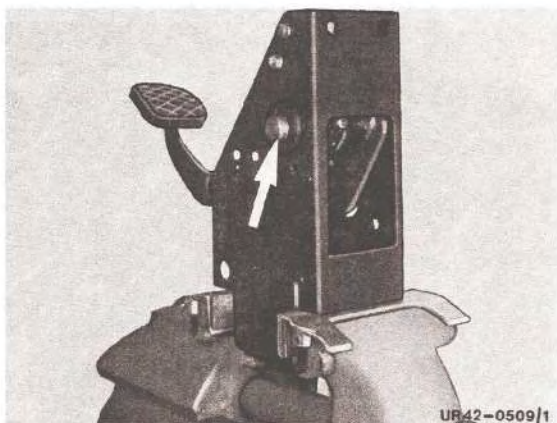
- 7 Attach cable to stoplight switch.

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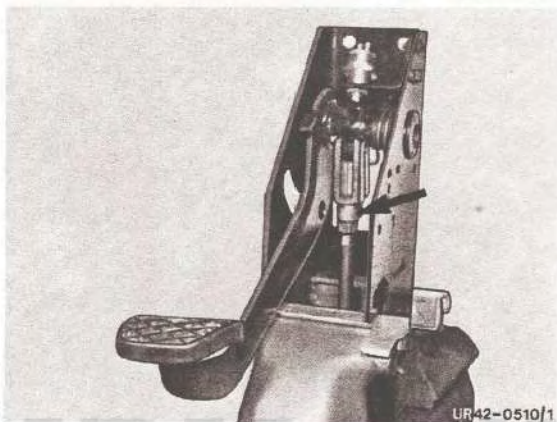
Removal, with linkage

- 1 Prop up engine hood.
- 2 Drain compressed air off tanks.
- 3 Unscrew and remove service brake valve.
- 4 Unscrew cable at stoplight switch and detach pedal bearing bracket.
- 5 Unhook return spring from pedal bearing bracket
- 6 Unscrew bearing pin retainer.

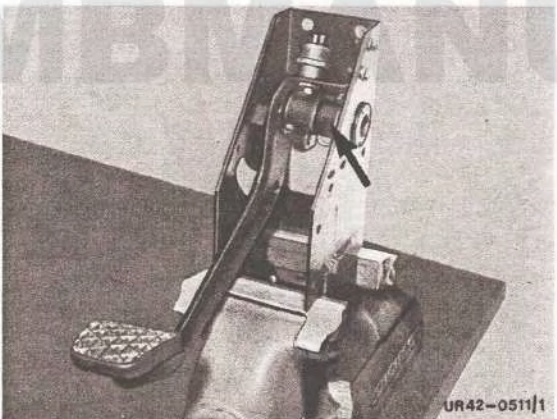




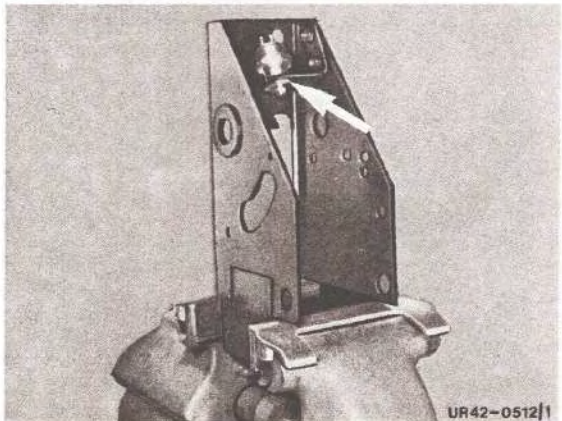
7 Remove bearing pin.



8 Detach piston rod retainer and remove piston rod.



9 Remove brake pedal.

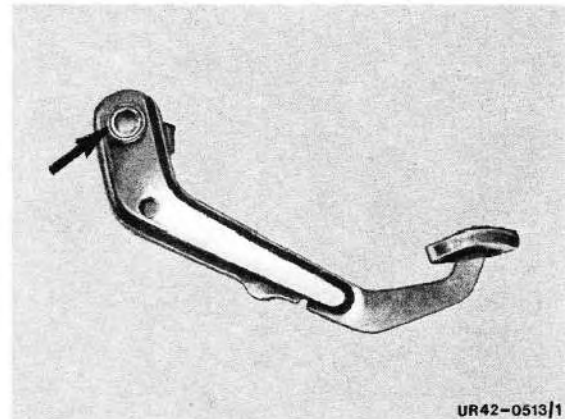


10 Unscrew stoplight switch.

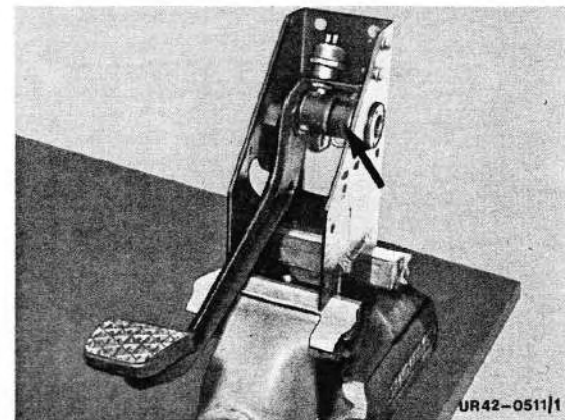
- 11 Check brake pedal bearing for wear, exchanging if necessary.
- 12 Clean and check all parts, exchanging if necessary.

Installation, with linkage

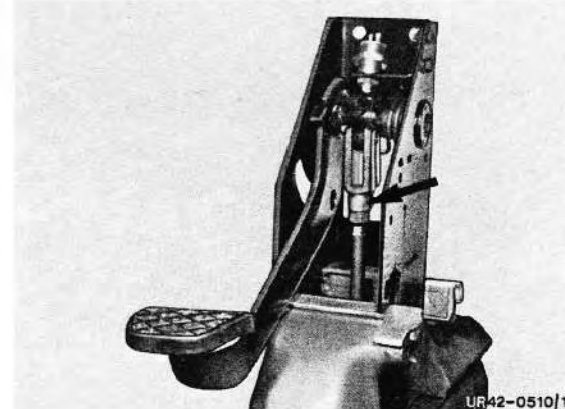
- 1 Screw stoplight switch on.
- 2 Fit brake pedal.
- 3 Fit piston rod and attach piston rod retainer.
- 4 Insert bearing pin.



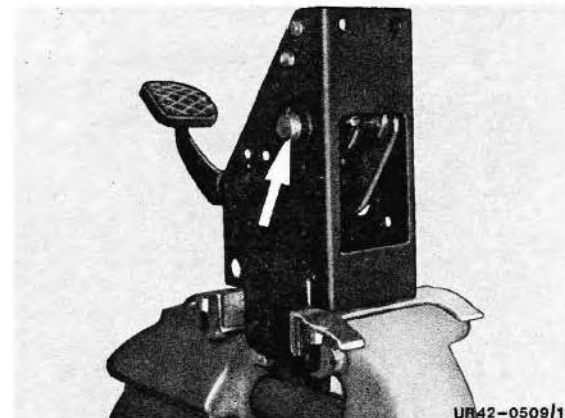
UR42-0513/1



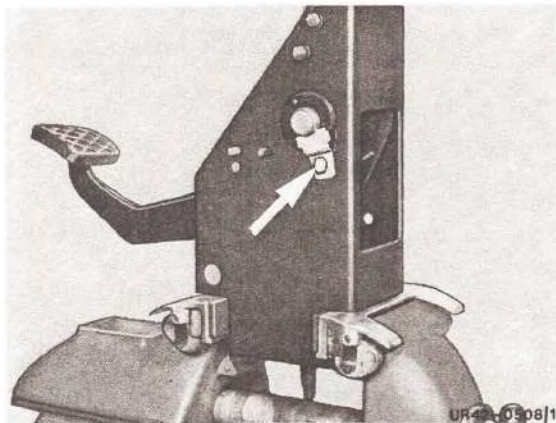
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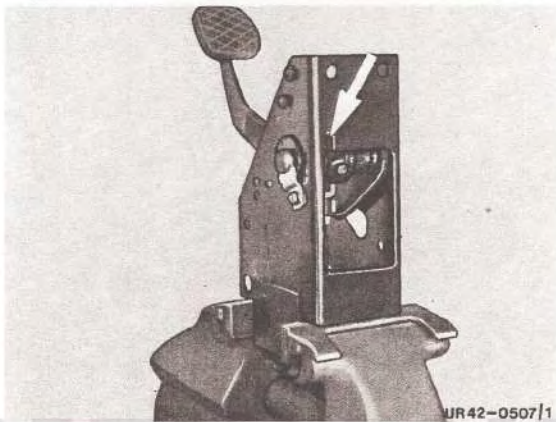
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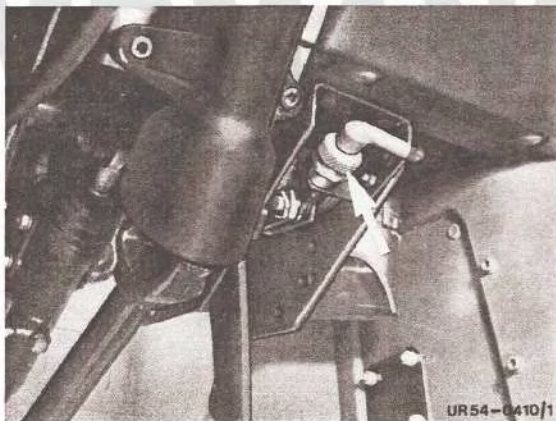
UR42-0509/1



5 Screw bearing pin retainer on.



6 Hook return spring onto pedal bearing bracket.

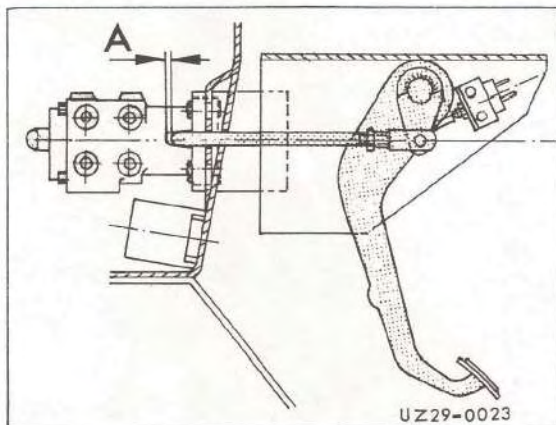


7 Attach pedal bearing bracket and screw stop-light switch on.

8 Install and fasten service brake valve.

9 Build up working pressure in air system.

10 Lower engine hood.



11 Check play "A" between piston rod and service brake valve, adjusting to **0.5 mm** if necessary.

Note: This amount of clearance cannot be measured and has to be felt at piston rod.

12 Check system for satisfactory operation.

Survey

Version

Control standard

30.1

Control SA 35 989

30.4

Contents

Chapter	Page
1 General	
Installation survey	1.1/1
Exploded view	1.2/1
2 Adjustment of linkage	2.1/1

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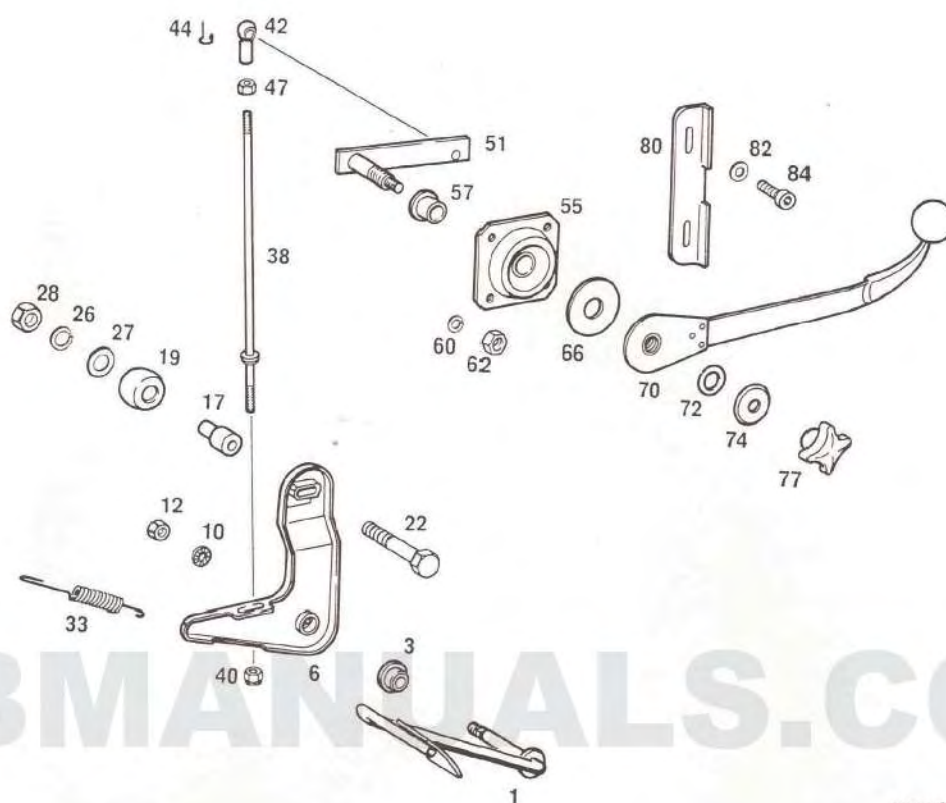
Installation survey

Chassis Model	Sales designation	Standard End chassis No.	Cab Model	Installation
435.110	U 1700	from 002 230	425.800 425.820 425.821 425.825	Standard from SP ²⁾
.111	U 1700 L ¹⁾	from 000 008		
.113	U 1700 L/38	from 084 642		
.115	U 1300 L ¹⁾	from 000 019		
.117	U 1300 L/37	from 071 796		

1) L = Lorry

2) SP = Start of production

Exploded view



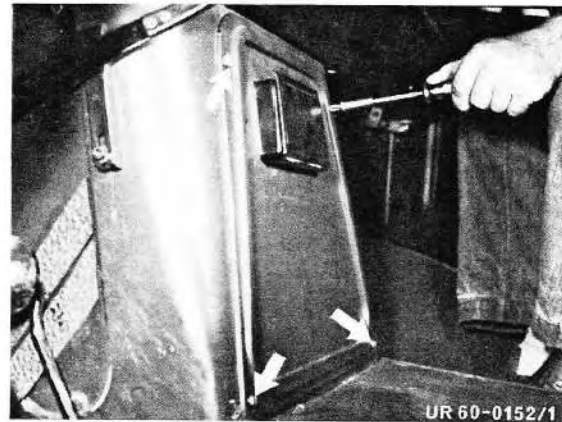
UZ30-0004

Control

- | | |
|---------------------|-----------------------|
| 1 Accelerator pedal | 44 Locking bar |
| 3 Bushing | 47 Nut |
| 6 Lever | 51 Lever |
| 10 Lock washer | 55 Housing |
| 12 Nut | 57 Bushing |
| 17 Spacing tube | 60 Lock washer |
| 19 Roller | 62 Nut |
| 22 Bolt | 66 Thrust washer |
| 26 Lock washer | 70 Hand control lever |
| 27 Washer | 72 Rubber buffer |
| 28 Nut | 74 Thrust washer |
| 33 Spring | 77 Start knob |
| 38 Control rod | 80 Idle detent |
| 40 Stop nut | 82 Washer |
| 42 Socket | 84 Bolt |

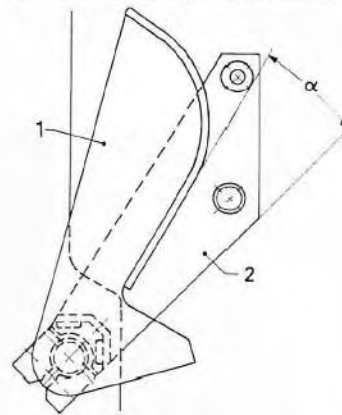
Adjustment

1 Detach engine cover in cab.



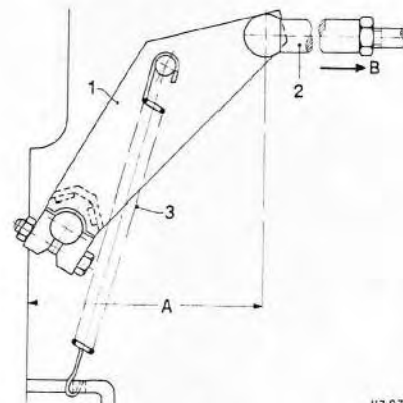
2 Slip lever onto intermediate spindle at right-hand side of engine so that the two levers (left and right) form an angle of about 15° .

- 1 Lever (left)
- 2 Lever (right)
- $\alpha = 15^\circ$



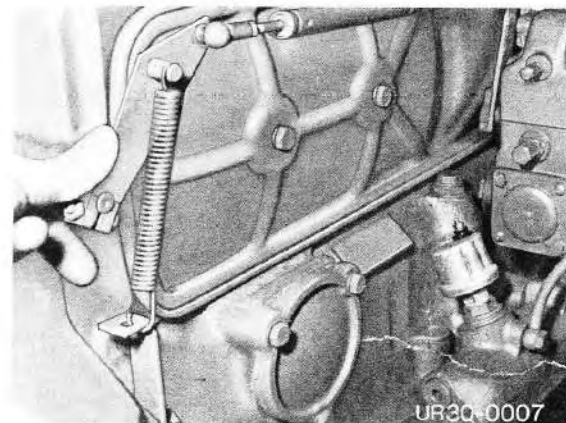
3 Adjust linkage between right-hand lever of intermediate spindle and injection pump control lever so that dimension A = 90 mm is maintained when control lever is moved to full load and spring is extended.

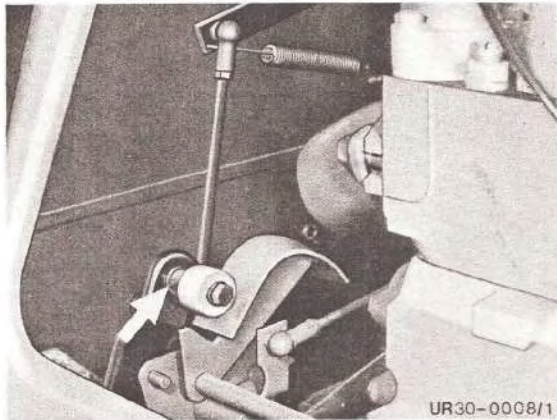
- 1 Lever
- 2 Pull rod
- 3 Spring
- A = 90 mm
- B = Control lever at full-load stop



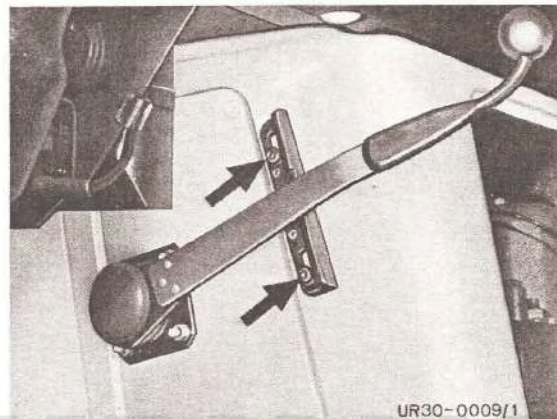
4 Force injection pump control lever to full load but do not compress spring in connecting linkage.

5 Press accelerator pedal down to within about 10 mm from floor of footwell.



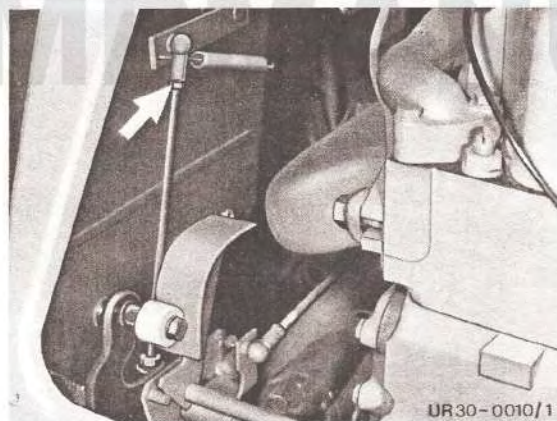


6 While maintaining this accelerator pedal position, adjust roller in oblong hole so that no play is left between roller and intermediate spindle lever.

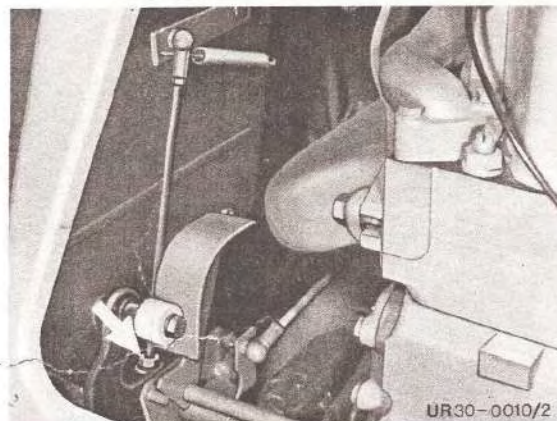


7 To adjust idle at hand control lever, release fastening screws and move idle stop accordingly.

Note: Moving upwards reduces engine speed.
Moving downwards accelerates engine speed.



8 If length of oblong hole in detent is inadequate, slacken lock-nut and then turn linkage between accelerator pedal and hand control lever to shorten or lengthen as required.



9 Adjust stop nut at bottom end of thrust linkage between hand control lever and accelerator pedal so that, on leaving stop position, hand control lever latches in idle detent as soon as accelerator pedal touches cab floor. (The spring actuator in linkage to pump control lever comes into action now.)

Note: In models featuring exhaust brake, do not adjust the stop nut.

Models incorporating an exhaust brake are to be adjusted as follows:

Execute jobs 1 to 6 in same manner as for engine without exhaust brake.

10 Having slackened lock-nut, adjust idling speed with aid of pull rod between intermediate lever and butterfly valve.

Note: Lengthening pull rod reduces engine speed. Shortening pull rod accelerates engine speed. Please note that exhaust brake cylinder is not fully retracted at this position (preloading).

11 Operate exhaust brake.

12 Undo screw M 6 at intermediate lever until clearance of 1 to 2 mm is obtained from intermediate spindle lever.

- 1 Intermediate starter flange
- 2 Exhaust brake cylinder
- 3 Pull rod, butterfly valve
- 4 Adjusting screw
- 5 Lever
- 6 Intermediate lever
- 7 Idle stop
- A = 1 to 2 mm

13 Attach engine cover in cab.

